Mir Sameen Hasnat

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EDUCATION

University of British Columbia, Canada — *CGPA: 86%*
B.A.Sc, Electrical Engineering (2nd Year) — *Expected Graduation: 2029*
- *Dean's List (Fall 2026)*

TECHNICAL SKILLS

Languages: C, C++, Java, Python, MATLAB
Hardware and Embedded Systems: Arduino, ESP32, RLC Circuits, PCB Design and Schematics
Fabrication: 3D Printing, Sheet Metal, CNC Machines, Soldering, Hand and Power Tools
Design: Fusion 360, SolidWorks, LTspice, Tinkercad, Adobe Creative Suite, Canva

TECHNICAL WORK EXPERIENCE

Electrical Team Member | *UBC Agrobots* — *Jan 2026 - present*
- Engineered an automated robot with refined power systems, run using an NVIDIA Jetson Orin Developer Kit for AI-controlled automation and ESP32 for motor and component control
- Researched efficient methods to integrate components with variable power ratings, powered by a 48V battery, using buck converters and bus bars while designing circuit schematics for PCB layout
- Designed the power distribution behind automated farming capabilities including weed control, seed sowing and crop harvesting across a plantation
- Collaborated with a team of engineers to develop the flagship Agrobot, maintaining documentation and coordination across 4 sub-teams

OTHER WORK EXPERIENCE

Production Worker | *Floform Industries* — *Jul 2026 – Aug 2026*
- Fabricated countertops by cutting to spec using CNC machines, table saws and staple guns to meet client deadlines
- Coordinated across 4 stations to produce countertops, often working multiple stations such as buildup, spray and lamination

Market Research Interviewer | *Access Research* — *Aug 2023 - Aug 2025*
- Conducted surveys on behalf of institutions such as colleges, Toronto Police Service, and various public and private sector clients, achieving a 92% communication and customer service rating
- Collected field research and data for media and communication development projects

Treasurer and Teaching Assistant | *MLIS Robotics Club* — *Aug 2022 - Jul 2023*
- Taught a year-long robotics course on Arduino IDE to 100+ students to increase technology accessibility in our community
- Organized a PC-building workshop sponsored by Corsair for 300+ students for brand development as well as greater tech enthusiasm
- Founded a startup selling robotics parts for the club, generating its first-ever profits of $2000, while also raising funding for club operations

TECHNICAL PROJECTS

Bladder Buddy | *C++, E-CAD, LTspice, ESP32* — *2026*
- *Special Recognition, UBC Design Team Competition (2026)*
- Designed automatic monitoring of urine content, including protein and pH levels, to diagnose health problems from public urinals
- Built self-designed protein and pH sensors using a GUVA UV sensor, pH probe, and disc motors
- Reduced pre-screening wait times from 2-3 business days to 2 minutes

Spider Bot | *C++, CAD, Arduino* — *2024*
- *3rd Place, WMC Annual Robotics Competition (2024)*
- Built a Bluetooth-controlled robot that replicates a spider's walk across four legs, navigating rough terrain under user control
- Achieved a compact, 150g build powered by a small LiPo battery for rapid, agile movement

Mars 2.0 | *Arduino IDE, C++, CAD Modelling, Tinkercad* — *2021*
- *1st Place, MLIS Annual Science Fair and 3rd Place, American Embassy Project Exhibition (2021)*
- Implemented motion and obstacle detection to safely navigate tight spaces in simulated earthquake scenarios
- Engineered a Bluetooth-enabled manual override with a servo-controlled camera, giving the user a live visual feed at a control range of 1.5 km

COMMUNITY PROJECTS

Founder | *Reboot* *2026 - present*

- Researched motherboard functionality and hardware fault diagnosis to develop efficient methods for turning scrapped laptops into functional machines
- Maintained performance and cosmetic standards so every device meets a usable baseline before it is given away
- Provided technology to underprivileged people to advance their education or careers, with each machine capable of simulation, modelling and productivity work
- Raised funding to buy and repair a wide variety of laptops